



Signals and Systems EE252, Spring 2026

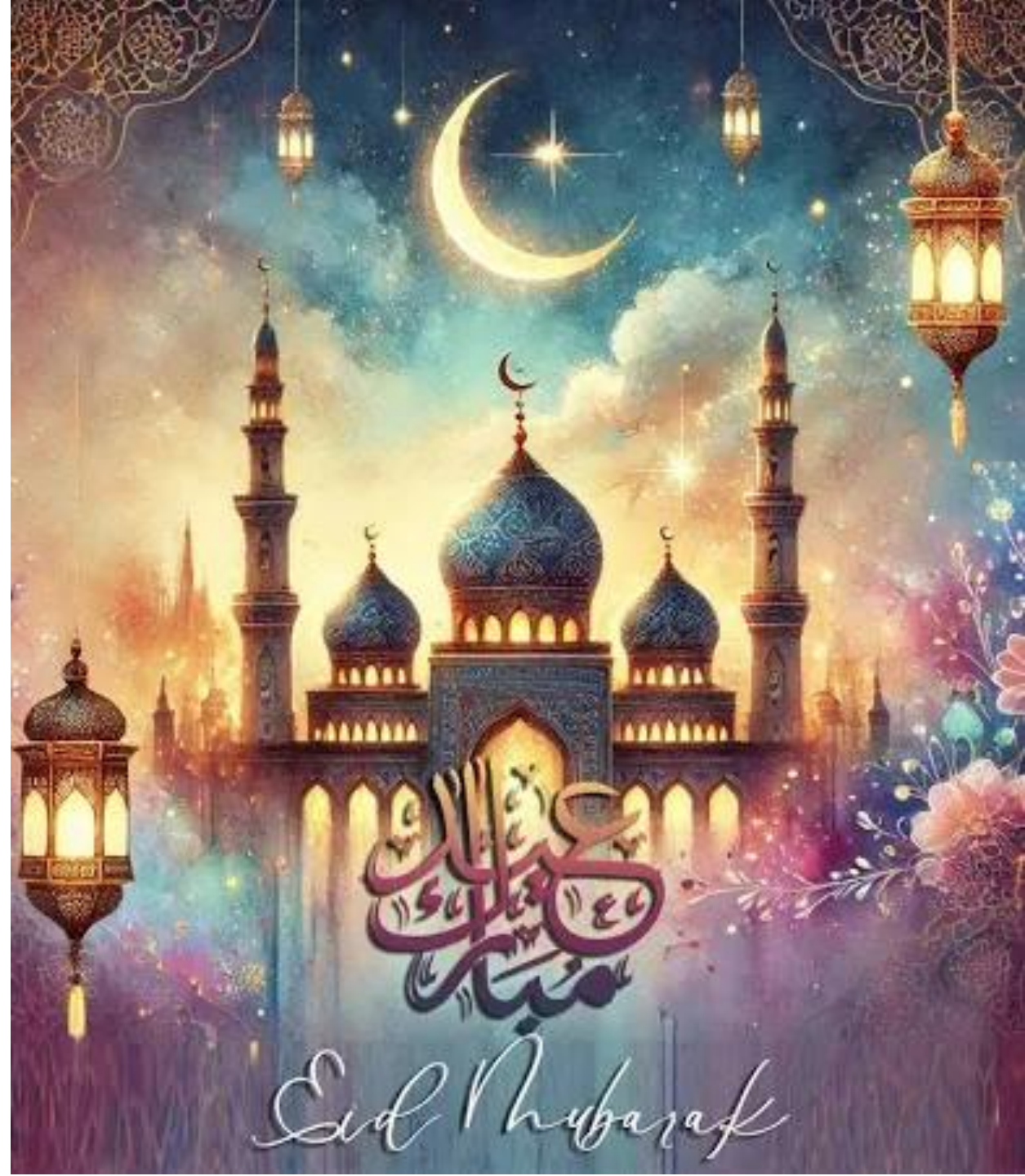
Instructor: Tariq Mumtaz

Semester week # 11



Welcome Back😊

Hope you had a nice Eid Break



Announcements:

- Homework-III is due over the weekend, online submission on LMS. Please read the assessment instructions carefully – 80% marks are based on solution steps.
- Quiz –III is scheduled on Monday, 03/30, during the routine class session.
- Project Milestone is due in Wk# 12 (coming week), each group has to lock 30 min slot for demo/viva session.



Course Learning Outcome (CLOs):

CLO-1	Apply knowledge of signals and systems to categorize and perform basic operations on signals and systems.	PLO-1 (Engineering Knowledge)	Cognitive Level-3 (Applying)
CLO-2	Calculate the response of DT and CT LTI systems to various inputs and categorize LTI systems.	PLO-1 (Engineering Knowledge)	Cognitive Level-3 (Applying)
CLO-3	Analyze DT signals and systems in the transform domain using the Z transform.	PLO-2 (Problem Analysis)	Cognitive Level-4 (Analyzing)
CLO-4	Analyze CT signals and systems in the transform domain using Laplace, Fourier transforms, and Fourier Series.	PLO-2 (Problem Analysis)	Cognitive Level-4 (Analyzing)
CLO-5	Investigate complex engineering problems in signals and systems and formally present the results.	PLO-4 (Investigation)	Cognitive Level-5 (Evaluating)



Roadmap till Semester end:

Frequency Domain Analysis of CT/DT Signals & Systems:

- ➔ Z- Transform (Wk 7 – 11)
- Laplace Transform (Wk 12 – 13)
- Fourier Transform (Wk 14 – 15)
- Project presentations (Wk16)



Learning outcome of last week's session:

Z- Transform solution of Difference Equation – 5.3

- Z Transform – Systems Perspective
- DT System Modeling from Difference Equations
- Solution of Difference Equations:
 - Time- domain Recursive/iterative method
 - DT System total response in the time domain– Zero input & Zero State response
 - Frequency-domain Z transform method
 - DT System total response in the frequency domain– Zero input & Zero State response



Learning outcome of today's session:

Z- Transform solution of Difference Equation – 5.3

- Z Transform – Systems Perspective
- DT System analysis:
 - Time-domain impulse response $h[n]$
 - Frequency-domain Transfer function $H[z]$
 - DT System Design using $H[z]$ and associated Poles & Zeros
 - Poles & Zeros – System stability and invertibility analysis



Z Transform: Systems Perspective

Time- Domain impulse response $h[n]$ & Frequency-Domain Transfer function $H[z]$



Z Transform:

In system theory, a signal $x[n]$ is called an eigenfunction of a system if, when it is applied as an input, the output is simply a scaled version of the same signal.

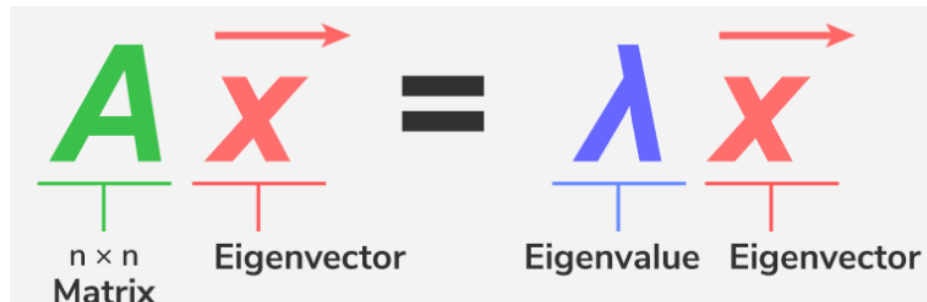
$$x[n] \rightarrow y[n] = \lambda x[n]$$

Input Output

Here, $x[n]$ is an eigenfunction and λ is the eigenvalue of the system.

Z Transform:

The Eigenvector and Eigenvalue definition:


$$\begin{array}{c} \text{A} \\ \hline n \times n \\ \text{Matrix} \end{array} \begin{array}{c} \xrightarrow{\hspace{1cm}} \\ \text{x} \\ \hline \text{Eigenvector} \end{array} = \begin{array}{c} \text{\lambda} \\ \hline \text{Eigenvalue} \end{array} \begin{array}{c} \xrightarrow{\hspace{1cm}} \\ \text{x} \\ \hline \text{Eigenvector} \end{array}$$

$A \rightarrow$ a linear operator (or matrix)

$x \rightarrow$ an eigenvector (or eigenfunction)

$\lambda \rightarrow$ eigenvalue (scalar)

Key benefit:

- *Instead of performing a complex and computationally extensive matrix multiplication, the operation reduces to a simple scaling by the eigenvalue λ .*
- *Where Scaling leads to a simple stretch or compression of the eigenvector.*

Applying the operator A to X does not change its direction/shape; rather, it scales it by λ .

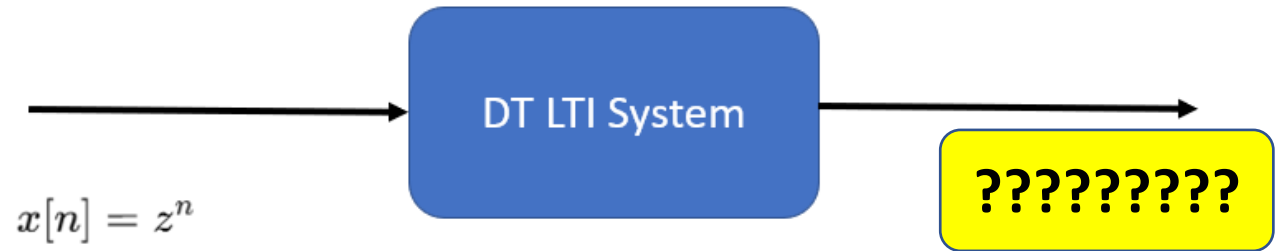
Z Transform:

$$y[n] = x[n] * h[n]$$

$$x[n] = z^n \quad \text{Input to the LTID system}$$

Output to the LTID system

$$y[n] = \sum_{k=-\infty}^{\infty} h[k] z^{n-k}$$



$$h[n] \longleftrightarrow H[z]$$

$$H(z) = \sum_{n=-\infty}^{\infty} h[n] z^{-n}$$

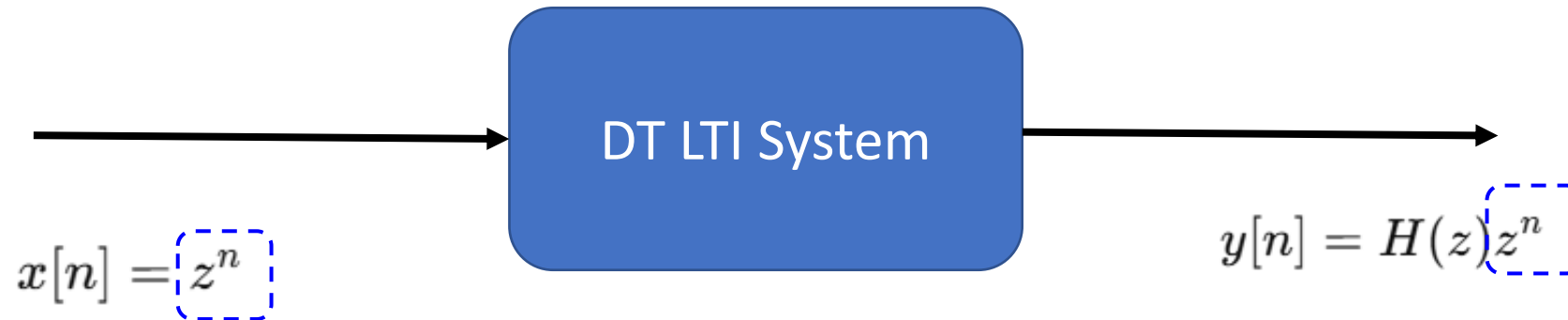
$$y[n] = z^n \sum_{k=-\infty}^{\infty} h[k] z^{-k}$$

$$y[n] = H(z) z^n$$

Here, the output of the LTID system is a scaled version of the input $x[n] = Z^n$
 That means, $x[n] = Z^n$ is the Eigenfunction while $H(Z)$ is the Eigenvalue

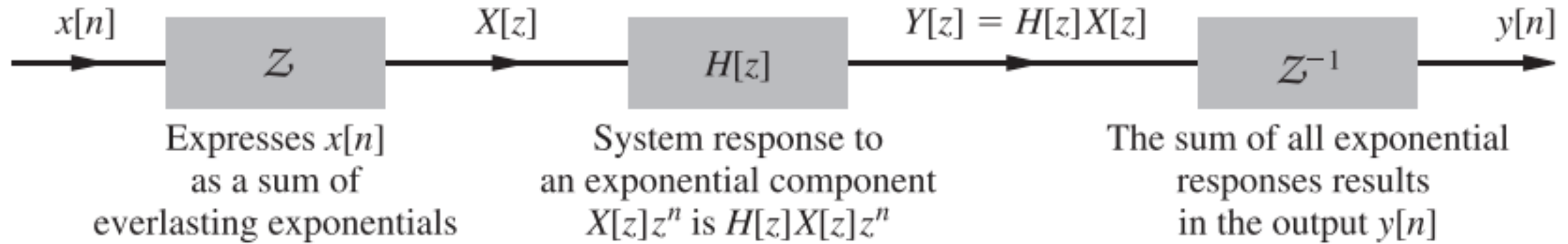
Z Transform:

The frequency-domain analysis of discrete-time systems is based on the fact that the response of a linear, time-invariant, discrete-time (LTID) system to an everlasting exponential z^n is the same exponential (within a multiplicative constant) given by $H[z]z^n$.

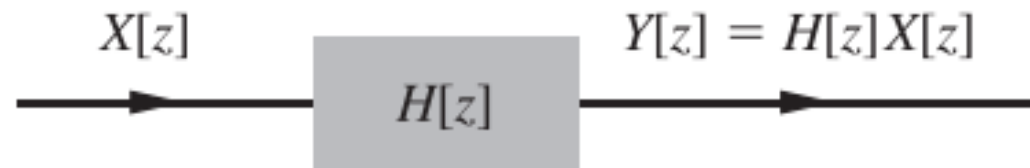


The output is the same signal scaled by $H(z)$

Z Transform:



(a)

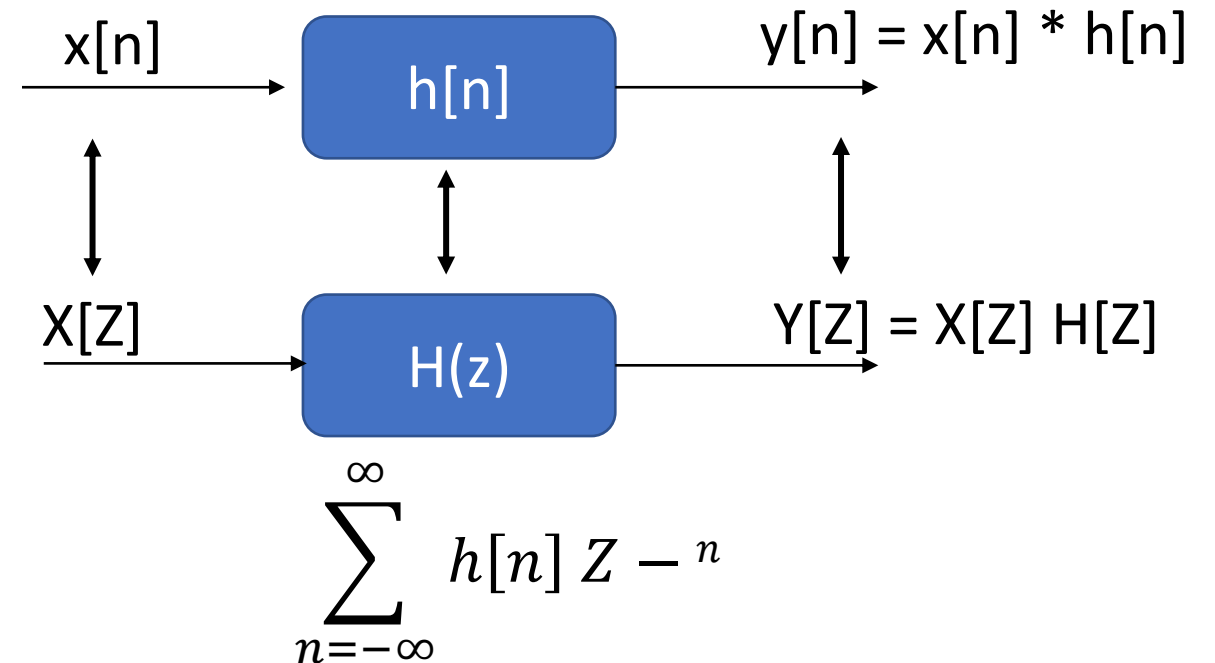


(b)

Z Transform:

Importance of Z Transform:

- It helps to find the output by a simple multiplication operation instead of time domain convolution.
- It helps analyze the system's stability and invertibility.



The z-transform $H(z)$ of $h[n]$ is referred to as the system function or the transfer function of the system.

$$H(z) = \frac{Y(z)}{X(z)}$$

Z Transform:

EXAMPLE 5.7 Transfer Function of a Unit Delay

Show that the transfer function of a unit delay is $1/z$.

If the input to the unit delay is $x[n]u[n]$, then its output (Fig. 5.7) is given by

$$y[n] = x[n - 1]u[n - 1]$$

The z -transform of this equation yields [see Eq. (5.12)]

$$Y[z] = \frac{1}{z}X[z] = H[z]X[z]$$

It follows that the transfer function of the unit delay is

$$H[z] = \frac{1}{z}$$

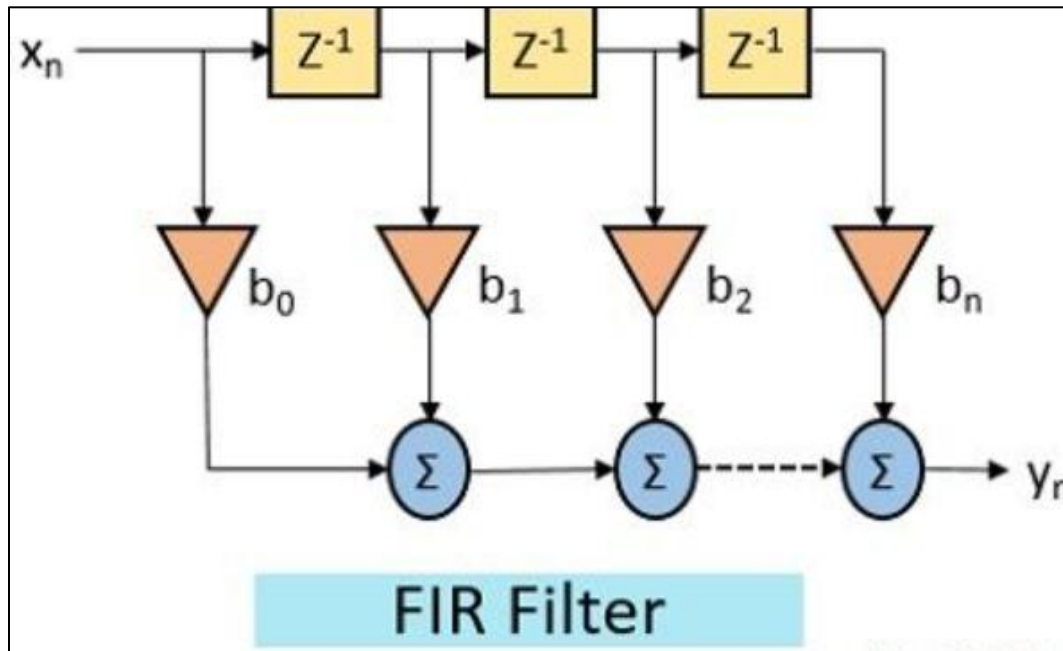


Figure 5.7 Ideal unit delay and its transfer function.

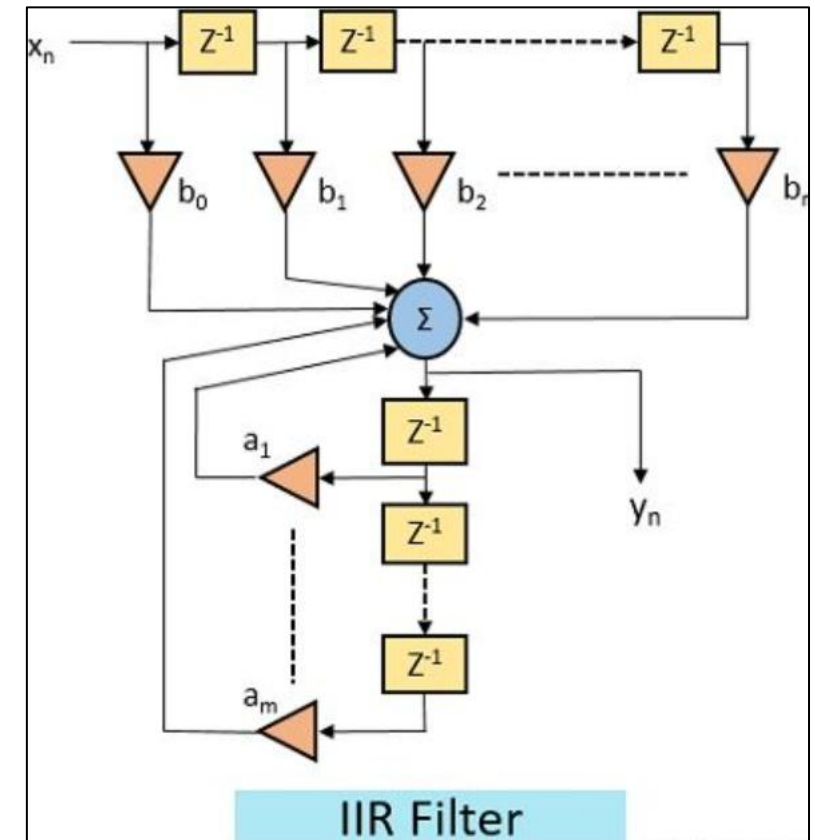
Z Transform:

Example: FIR (Finite Impulse Response) system

Here delay $x[n-1]$ is modeled as Z^{-1}



Example: IIR (Infinite Impulse Response) system



The Z Transform- System Analysis

$H(z)$ Transfer function- Poles & Zeros

Z Transform:

$$y[n] = -a_1y[n-1] - a_2y[n-2] - \dots - a_Ny[n-N] \\ + b_0x[n] + b_1x[n-1] + \dots + b_Nx[n-N]$$

Using the time-shift property and assuming zero initial conditions...

$$\left(1 + \frac{a_1}{z} + \frac{a_2}{z^2} + \dots + \frac{a_N}{z^N}\right) Y[z] = \left(b_0 + \frac{b_1}{z} + \frac{b_2}{z^2} + \dots + \frac{b_N}{z^N}\right) X[z]$$

$$Y[z] = \left(\frac{b_0z^N + b_1z^{N-1} + \dots + b_{N-1}z + b_N}{z^N + a_1z^{N-1} + \dots + a_{N-1}z + a_N}\right) X[z] \\ = \frac{P[z]}{Q[z]} X[z]$$

$$Y[z] = X[z]H[z]$$

$$H(z) = \frac{P(z)}{Q(z)}$$

$$y[n-m]u[n] \iff \frac{1}{z^m} Y[z]$$

$$x[n-m]u[n] \iff \frac{1}{z^m} X[z] \quad m = 1, 2, \dots, N$$

Z Transform:

The Transfer Function of a System:

$$H[z] = \frac{P[z]}{Q[z]} = \frac{b_0 z^N + b_1 z^{N-1} + \dots + b_{N-1} z + b_N}{z^N + a_1 z^{N-1} + \dots + a_{N-1} z + a_N}$$

$$H(z) = \frac{P(z)}{Q(z)} \quad \begin{array}{l} P(z) \rightarrow \text{numerator polynomial} \rightarrow \text{zeros} \\ Q(z) \rightarrow \text{denominator polynomial} \rightarrow \text{poles} \end{array}$$

The factorization of the polynomials

$$P(z) = b_0(z - z_1)(z - z_2) \cdots (z - z_N)$$

$$Q(z) = (z - p_1)(z - p_2) \cdots (z - p_N)$$

$$H(z) = b_0 \frac{(z - z_1)(z - z_2) \cdots (z - z_N)}{(z - p_1)(z - p_2) \cdots (z - p_N)}$$



Z Transform:

$$H(z) = \frac{P(z)}{Q(z)}$$

$P(z) \rightarrow$ numerator polynomial $\rightarrow$ zeros

$Q(z) \rightarrow$ denominator polynomial $\rightarrow$ poles

$$H(z) = b_0 \frac{(z - z_1)(z - z_2) \cdots (z - z_N)}{(z - p_1)(z - p_2) \cdots (z - p_N)}$$

Zeros: They are the roots of the numerator of $H(z)$. At these points, the system completely attenuates the signal. $H(z) = 0$ @ Zeros

Poles: They are the roots of the denominator of $H(z)$. At these points, the system amplifies the signal or the system response becomes unbounded. $H(z) \rightarrow \infty$ @ Poles

Z Transform:

$$H(z) = b_0 \frac{(z - z_1)(z - z_2) \cdots (z - z_N)}{(z - p_1)(z - p_2) \cdots (z - p_N)}$$

- Roots ($z_1, p_1, \dots$) of the numerator and the denominator of $H(z)$ are among the most important design & analysis parameters used in modern engineering.
- Although this topic will be explored in depth in advanced Signals & Systems courses such as DSP, let's develop a brief understanding of a system's frequency response using the poles and zeros of $H(z)$.



Z Transform:

$$H[z] = \frac{7z - 6}{z^2 - z - 6}$$

$$H[z] = \frac{1}{7} \frac{z - 6/7}{(z + 2)(z - 3)}$$

- Zeros: $z = \frac{6}{7}$
- Poles: $z = -2, z = 3$

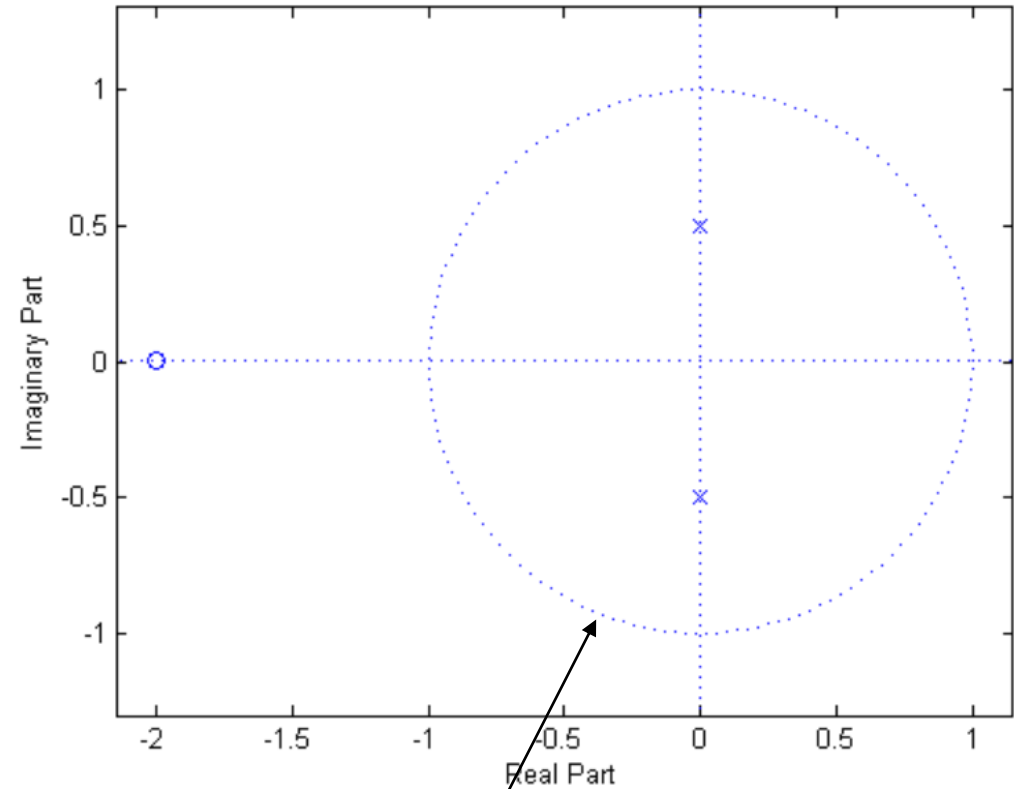
Z Transform:

We often find it useful to plot poles and zeros on the complex Z -plane with

- Poles marked as $\times$
- Zeros marked as 0

$$H(z) = \frac{z + 2}{z^2 + \frac{1}{4}}$$

- Zeros: $z = -2$
- Poles: $z = 0.5j, z = -0.5j$



Unit circle marked for reference



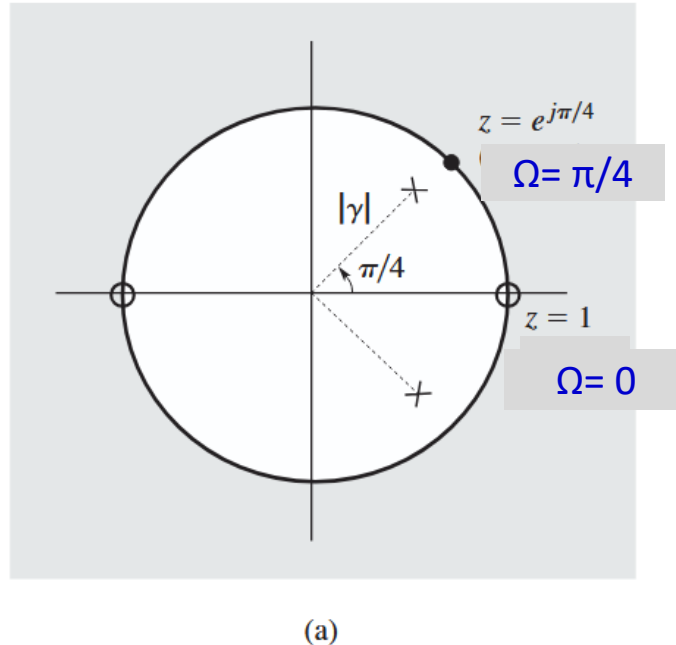
Z Transform:

The locations of Poles, Zeros, and their numbers play a critical role in understanding the response of a system to different frequencies.

- **Zeros control dips:** giving locations of frequencies that may be attenuated and how strong the attenuation will be.
- **Poles control peaks:** giving locations of frequencies that may be amplified and how strong the amplification will be.
- **Poles and zeros work together** to create a complete system response to different frequencies. They can reinforce or cancel each other's effect.

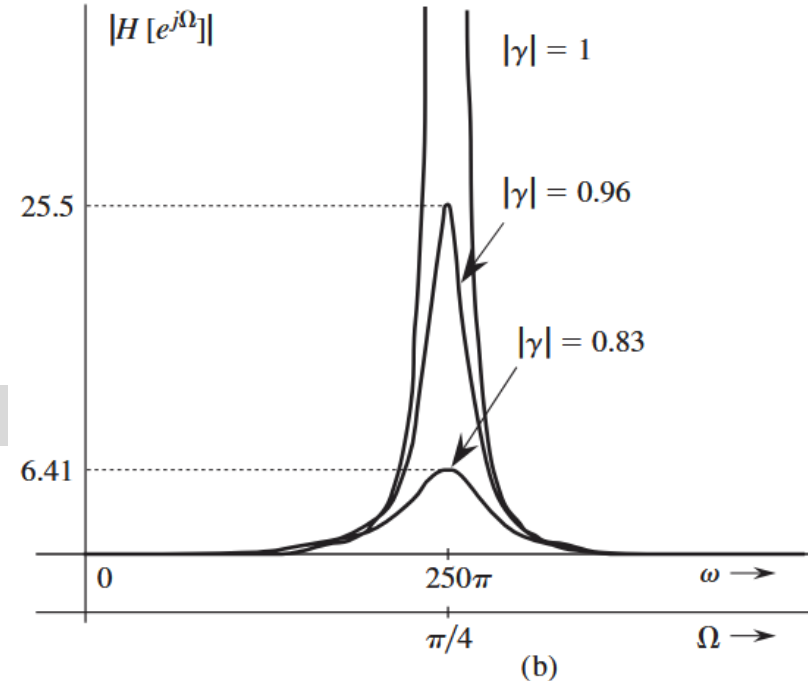
Z Transform

$$Z = \gamma e^{-j\Omega}$$



Poles Impact:

- This system says, I will amplify sinusoids of frequency $\pm \pi/4$ radians if they are in the input.
- The amplification amount will increase as the poles move away from the origin towards the unit circle.
- If they reach or cross the unit circle, the system will become unstable.



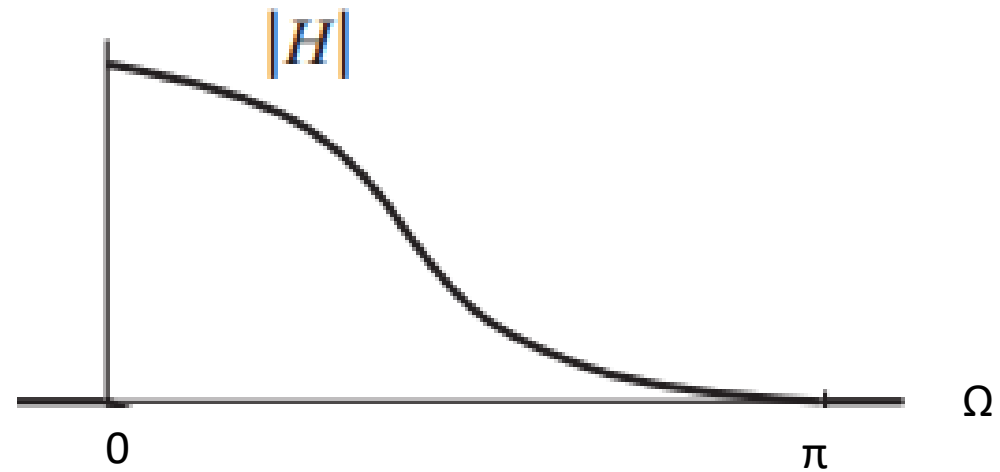
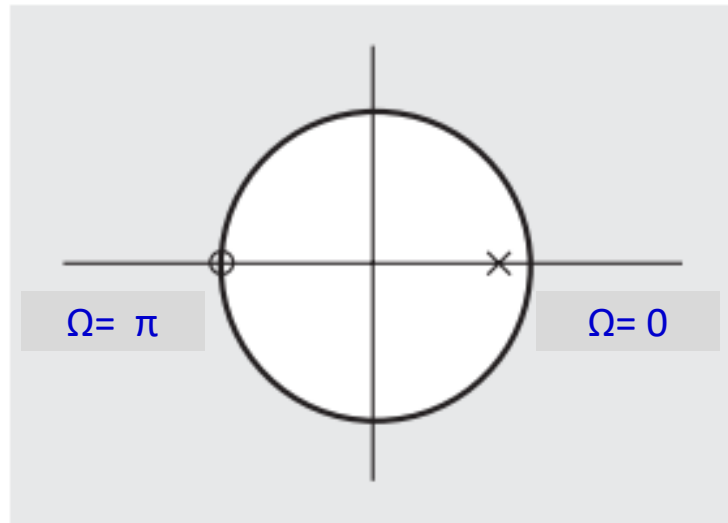
Zeros Impact:

- Zeros at 0 and π radians say that frequencies at and near these will be attenuated.
- Together, the poles and zeros form this response.

$\Omega = \omega T$
Here $T = 1000$

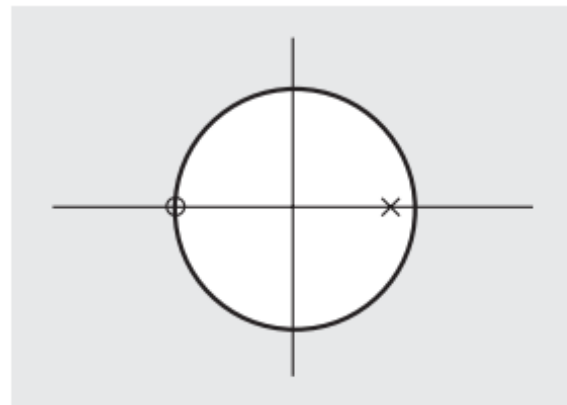
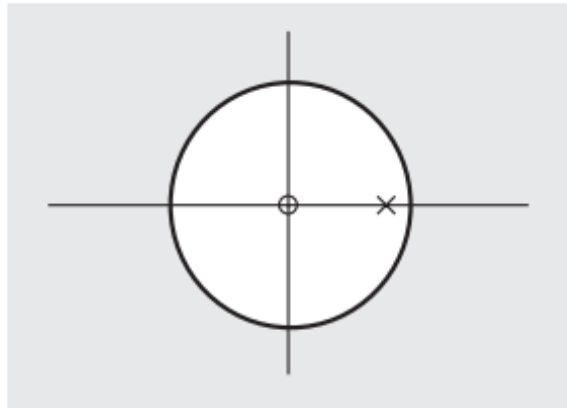
Z Transform:

Examples: First order Low pass filter

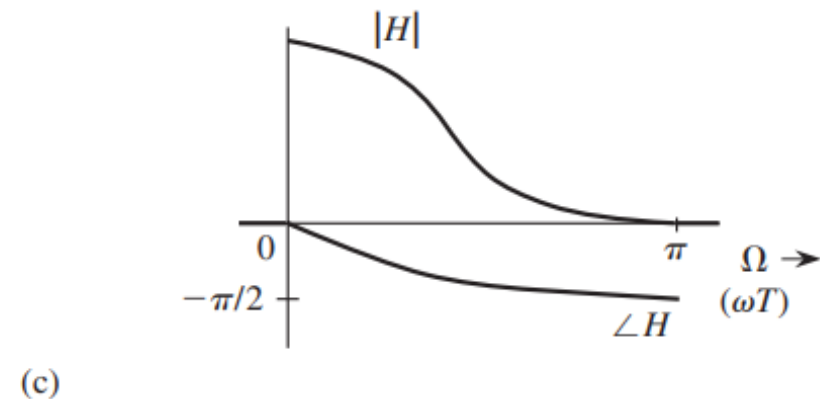
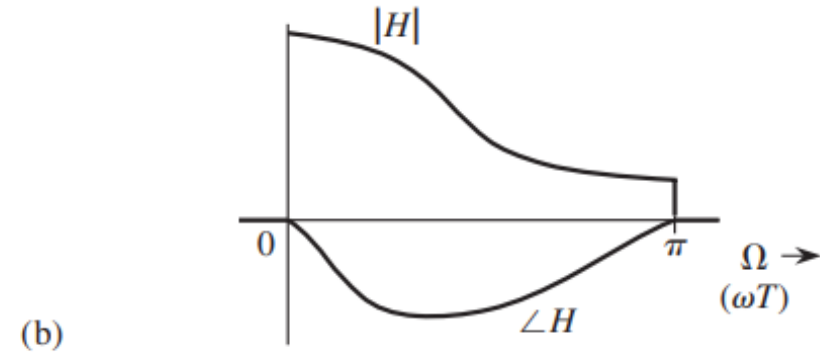


Z Transform:

Examples: First order Low pass filter

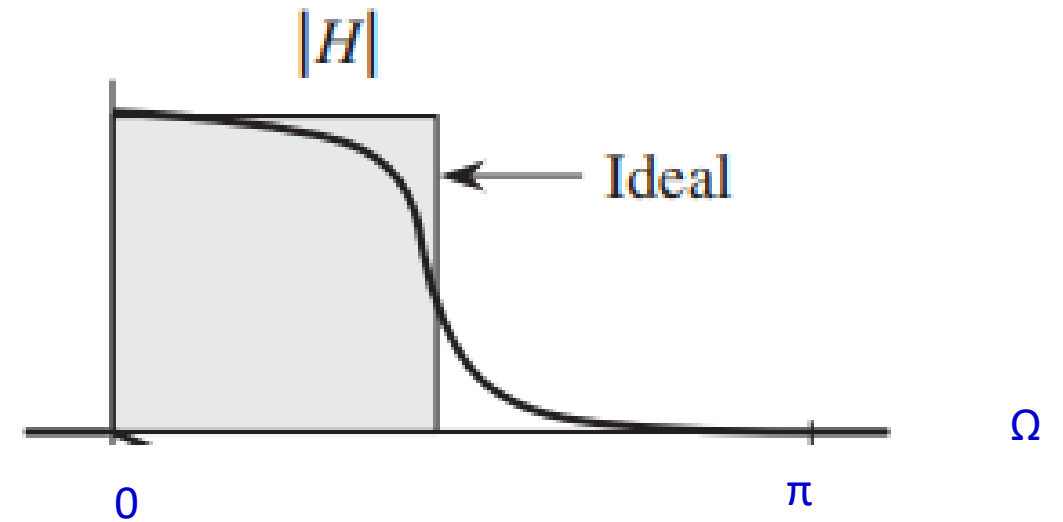
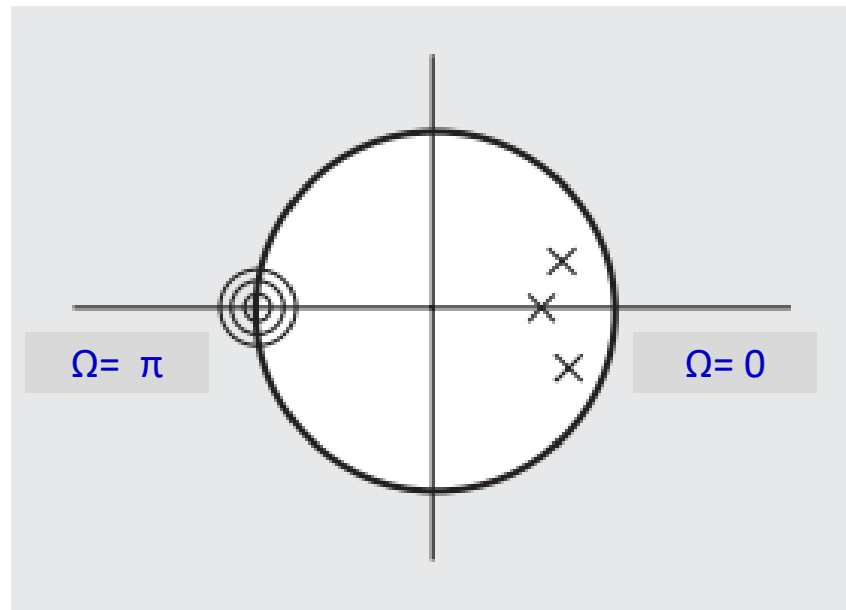


Impact of the placement of Zero



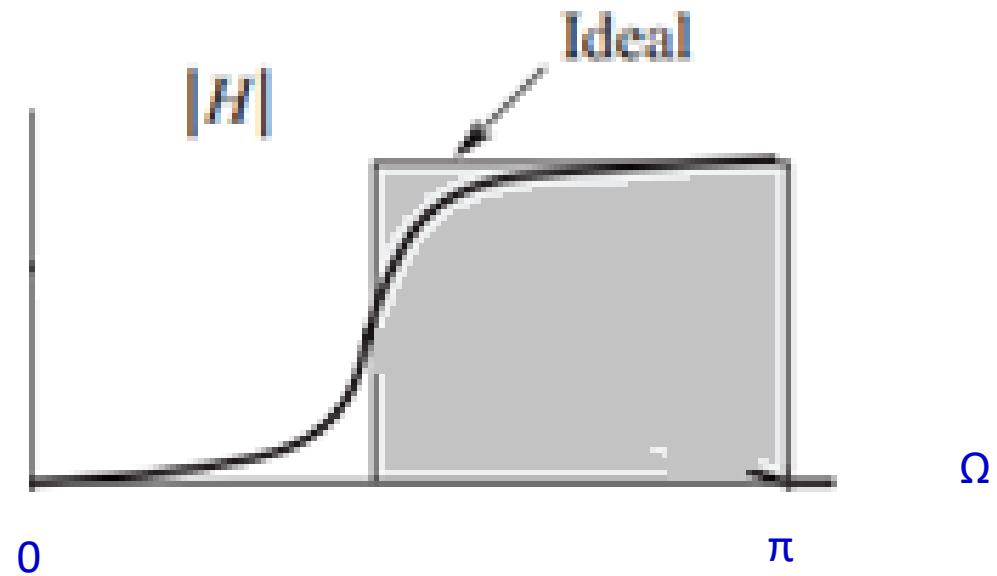
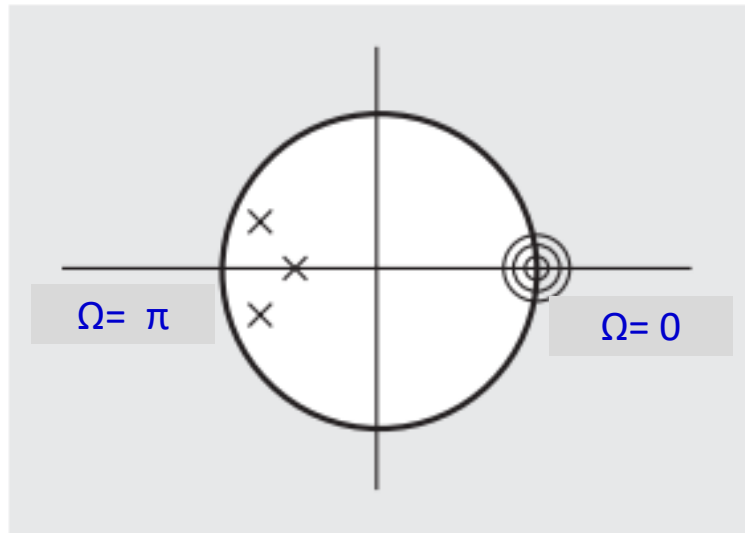
Z Transform:

Examples: High order Low pass filter



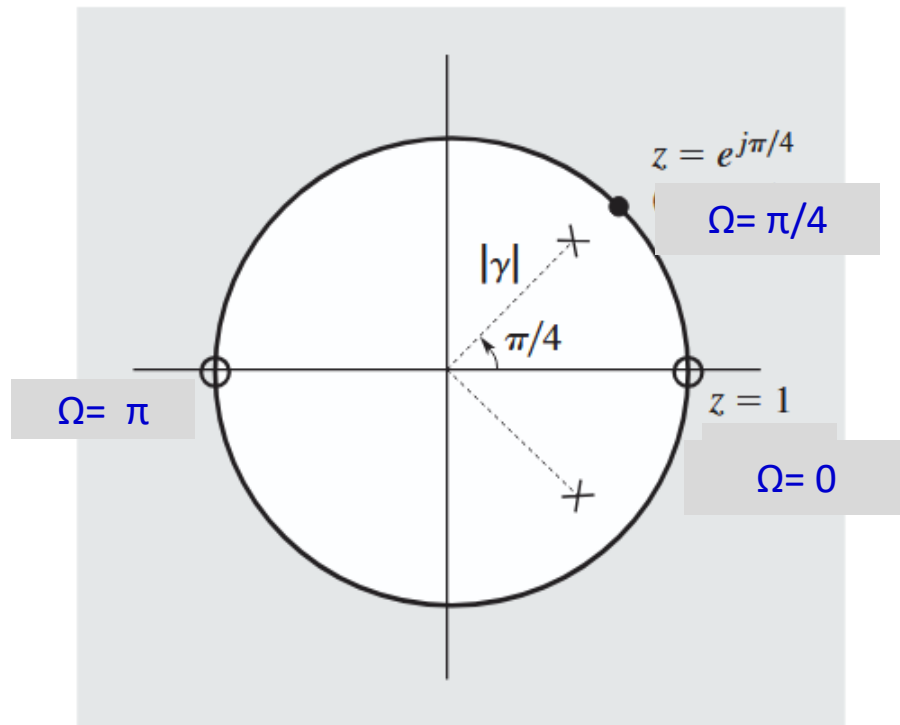
Z Transform:

Examples: High order High-pass filter

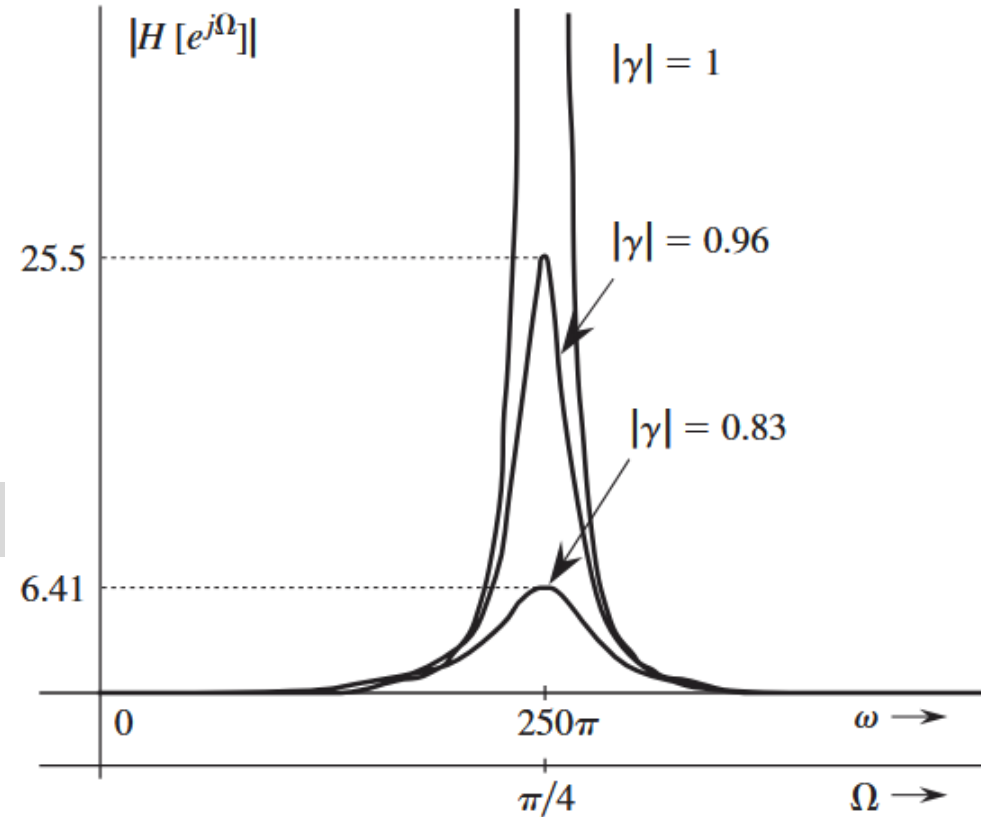


Z Transform:

Examples: High-order Bandpass filter



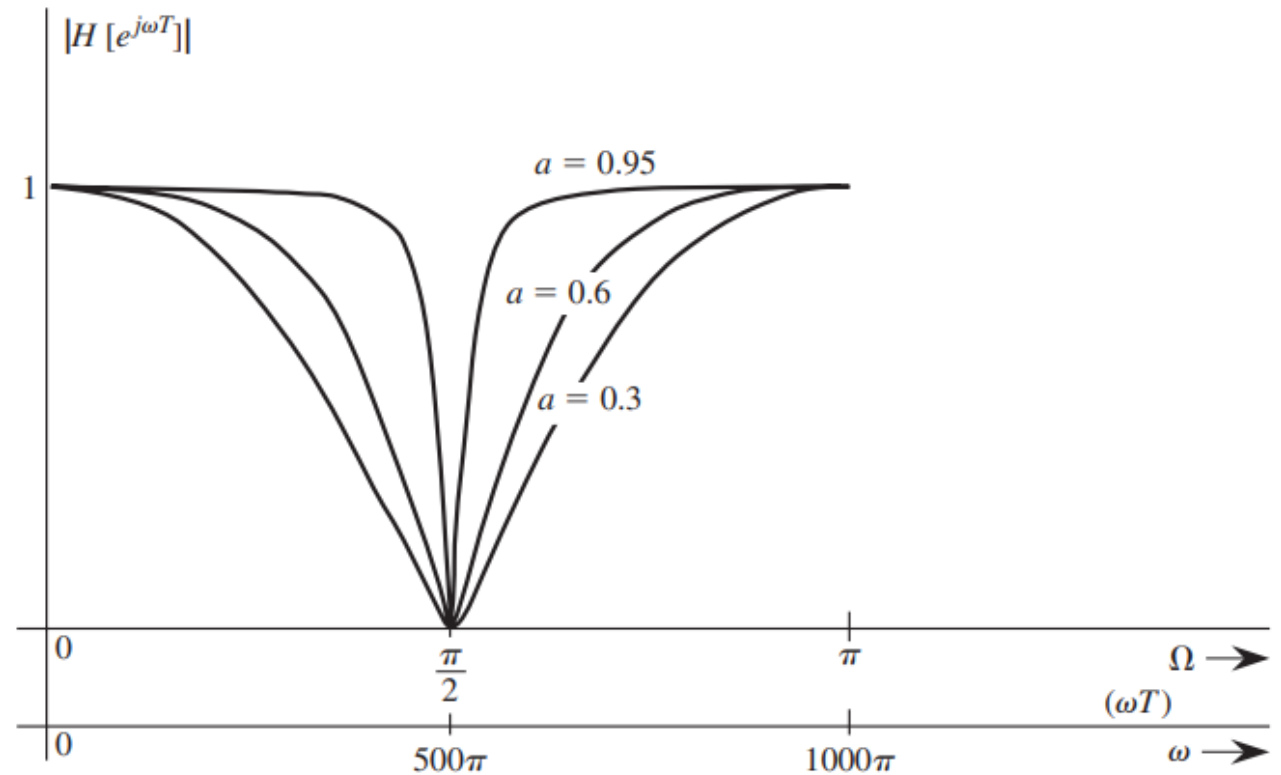
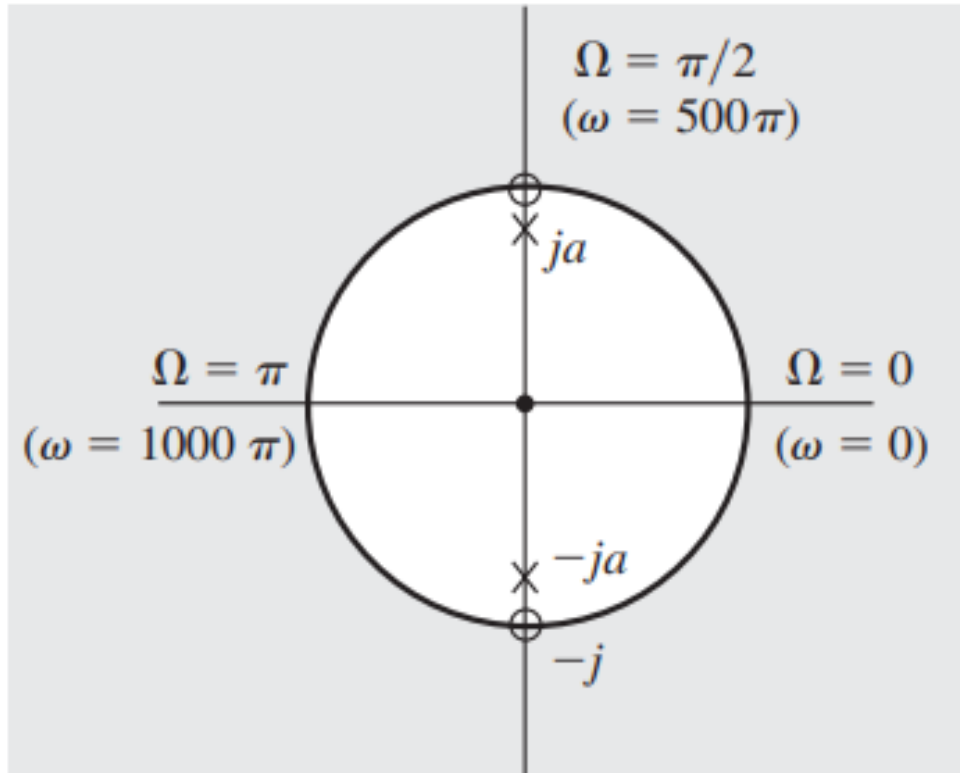
(a)



(b)

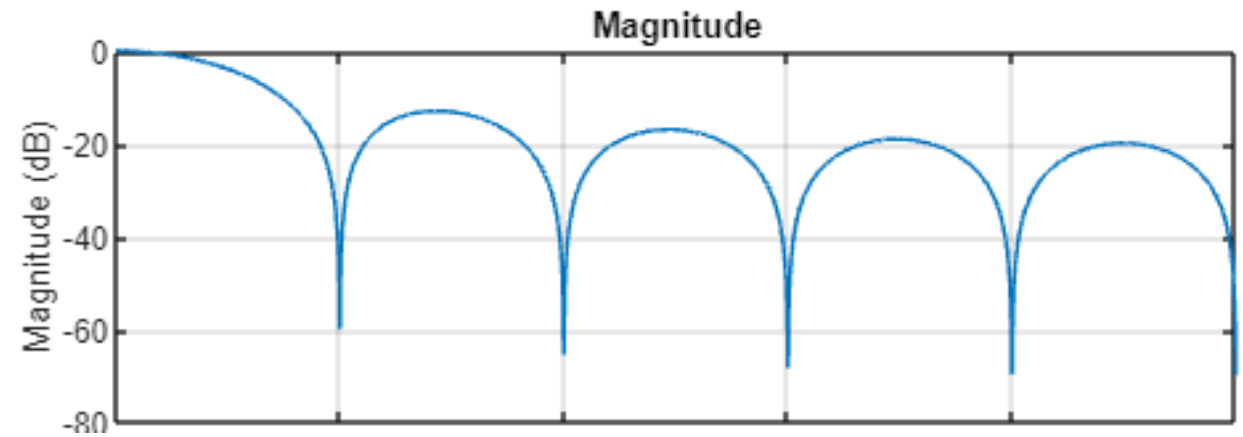
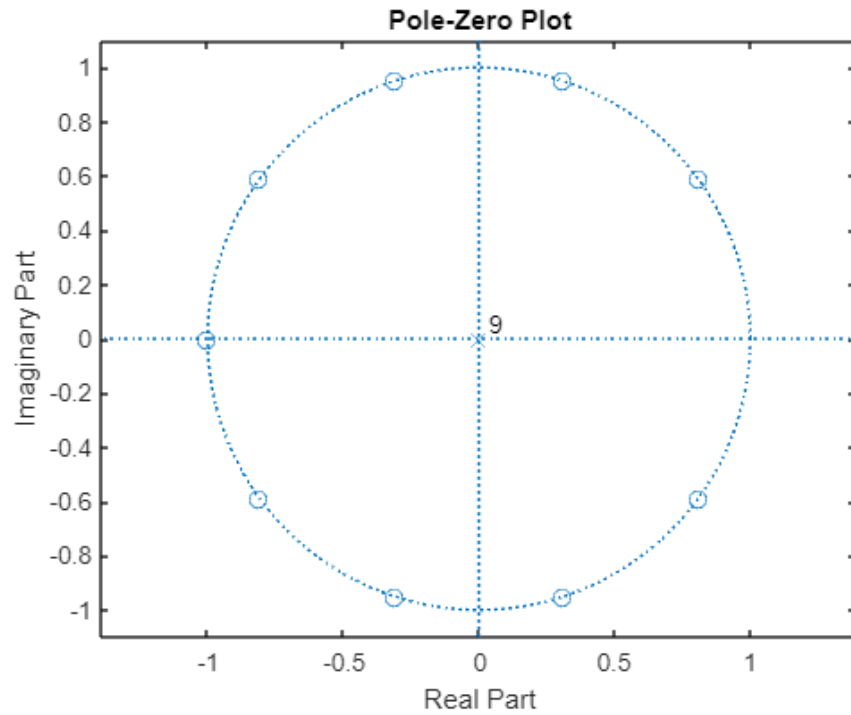
Z Transform:

Examples: High-order Notch filter



Z Transform:

Examples: High-order Comb filter



Z Transform:

[Applied DSP No. 9: The z-Domain and Parametric Filter Design - YouTube](#)



The Z Transform- System Analysis

Characterization of Discrete-Time LTI Systems



Z Transform:

Characterization of Discrete-Time LTI Systems through $H(Z)$:

The properties of discrete-time LTI systems are closely associated with the characteristics of $H(z)$ in the z -plane

- Stability
- Invertibility



Z Transform:

BIBO Stability:

A DT-LTI system is BIBO stable if all the poles of $H[z]$ are inside the unit circle i.e., for every pole z , we have $|z| < 1$

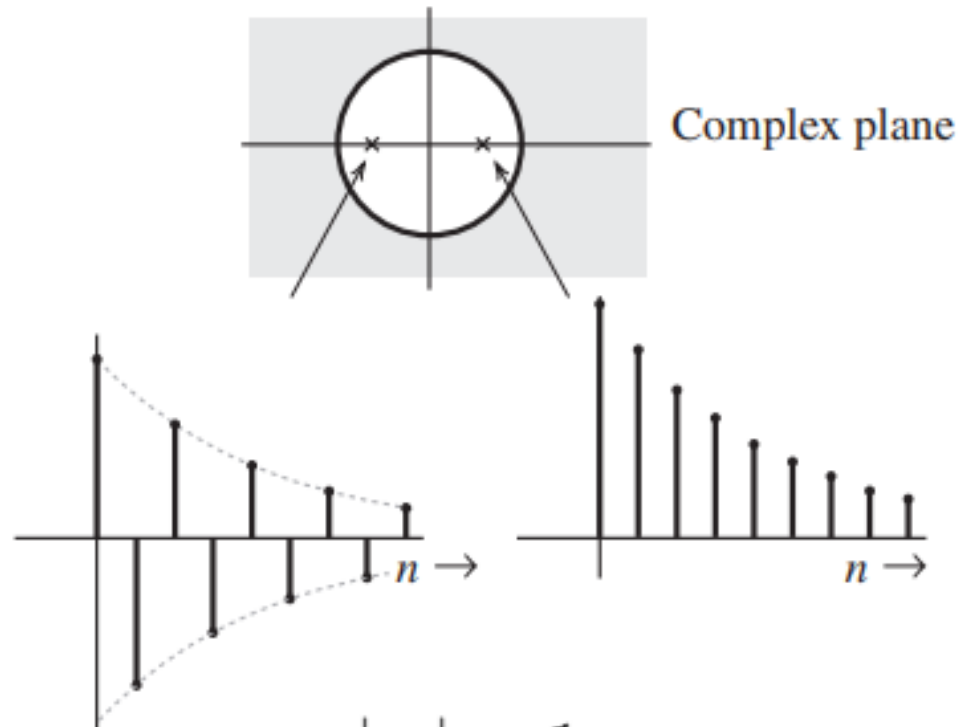
Reason:

- As we know $\gamma^n u[n] \Leftrightarrow \frac{z}{z-\gamma}$
- And the fact that BIBO stability requires $\sum |h[n]| < \infty$

- Notice that a pole (γ) in z becomes base of a geometric series in time.
- Clearly corresponding geometric series will be convergent $|\gamma| < 1$,
- This is equivalent to saying that pole is inside the unit circle (on complex plane).

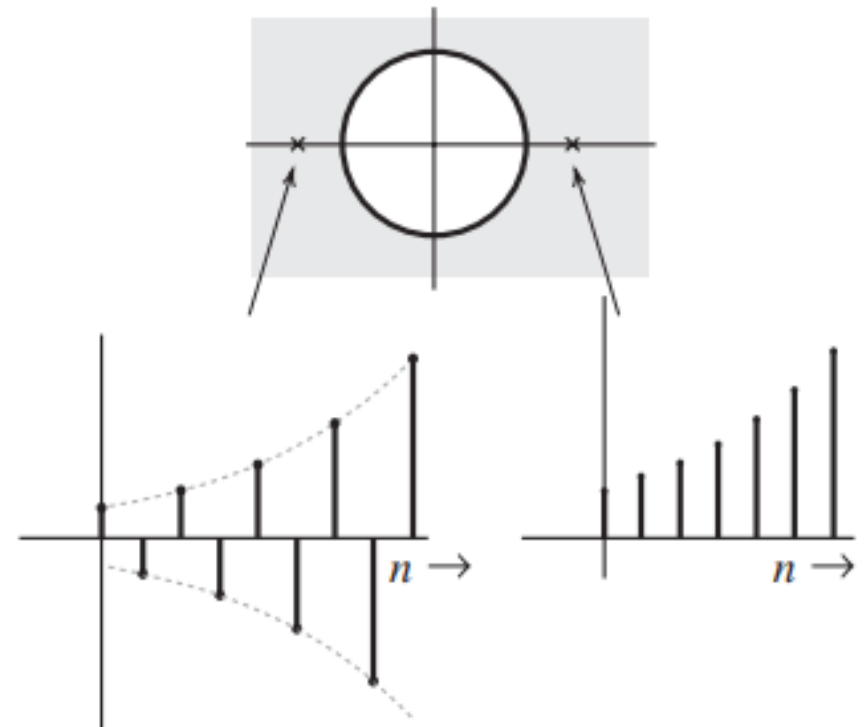
Z Transform: Stability

$$|p_i| < 1$$



$|p_i| < 1$ all the poles must be inside the unit circle for a BIBO Stable system

$$|p_i| > 1$$



Z Transform:

Example 1

$$H[z] = \frac{z}{z-2} \Leftrightarrow 2^n u[n] = h[n]$$

Pole outside
unit circle

Divergent
series

System BIBO unstable!

Example 2

$$H[z] = \frac{z}{z-0.5} \Leftrightarrow (0.5)^n u[n] = h[n]$$

Pole outside
inside unit circle

Convergent
series

System BIBO stable!



Z Transform:

Example 3

$$H[z] = \frac{z}{z-1} \Leftrightarrow (1)^n u[n] = h[n]$$

Pole outside
inside unit circle

Divergent
series

System BIBO unstable!

Pole on the unit circle is also BIBO unstable!



Z Transform:

- In general $H[z]$ can have multiple poles.
- Recall from inverse Z Tx, that we can use PFE to rewrite $H[z]$ such that poles are in separate terms!

Example 4

$$H[z] = \frac{8z - 19}{(z - 0.2)(z - 3)} = \frac{az}{z - 0.2} + \frac{bz}{z - 3}$$

$$h[n] = a(0.2)^n u[n] + b(3)^n u[n]$$

Convergent Divergent

Thus, even if one pole is on or outside the unit circle, it will lead to a divergent series $\mathbf{h[n]}$!



Z Transform:

Invertible:

A DT-LTI system is invertible if $H^{-1}(z) = \frac{1}{H(z)}$

- For this, a DT-LTI system is invertible if the numerator of $H[z]$ is not zero for any value of z .
- Because, if $H[z] = 0 \forall z$, then $Y[z] = H[z]X[z] = 0 \forall z$, and there is no way to recover $X[z]$.



Z Transform:

Stable Invertible

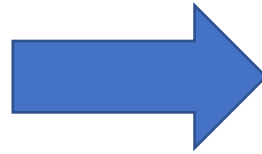
A DT-LTI system is stable invertible if all zeros of $H[z]$ are inside the unit circle.

Reason:

- Since $Y[z] = H[z]X[z]$, we need $H^{-1}[z]$ to recover $X[z]$ as $X[z] = H^{-1}[z]Y[z]$.
- But when we invert $H[z]$, the zeros become the poles of the inverse.
- Thus the inverse, $H^{-1}[z]$, will be stable only if all the zeros of $H[z]$ are inside the unit circle.

Z Transform:

$$H[z] = \frac{(z - 0.5)(z + 3)}{(z + 0.2)}$$



$$G[z] = H^{-1}[z] = \frac{(z + 0.2)}{(z - 0.5)(z + 3)}$$

Not stable-invertible, since one of the zeros ($z = -3$) is outside the unit circle.

Easy to see that the inverse is an unstable system with a pole ($z = -3$) outside the unit circle.

Since we are interested in system inverses that are stable, i.e. stable invertible.

The Z Transform

$H(z)$ Real world Application

Z Transform: Invertibility

If $H[z]$ is the transfer function of a system S , then S_i , its inverse system, has a transfer function $H_i[z]$ given by

$$H_i[z] = \frac{1}{H[z]}$$

If $H[z]$ is placed in cascade with $H_i[z]$, the transfer function of the composite system (identity system) is unity.



Z Transform: Invertibility



Example

$$H[z] = \frac{z - 0.4}{z - 0.7}$$

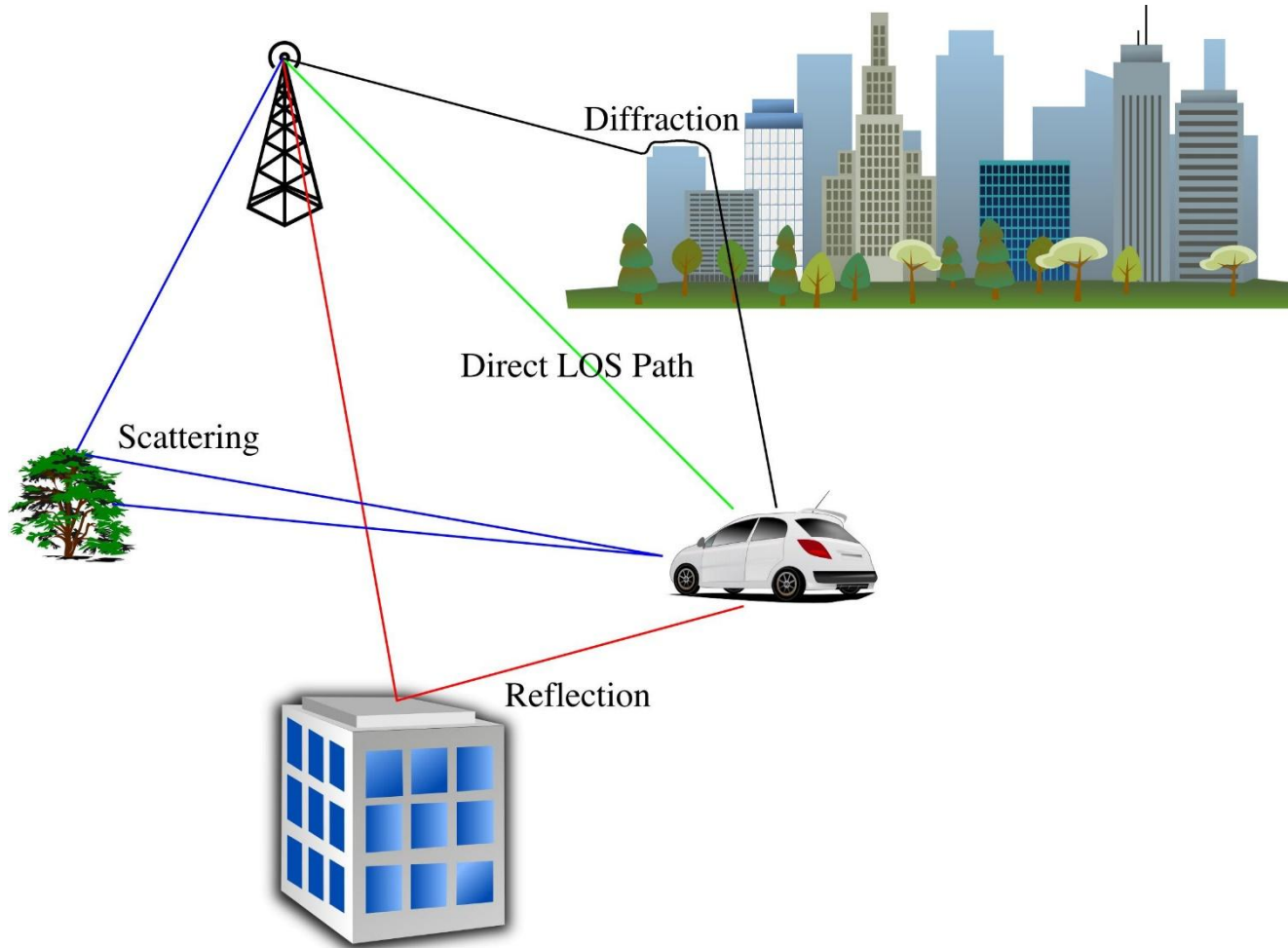
its inverse system transfer function is

$$H_i[z] = \frac{z - 0.7}{z - 0.4}$$

as required by the property $H[z]H_i[z] = 1$. Hence, it follows that

$$h[n] * h_i[n] = \delta[n]$$

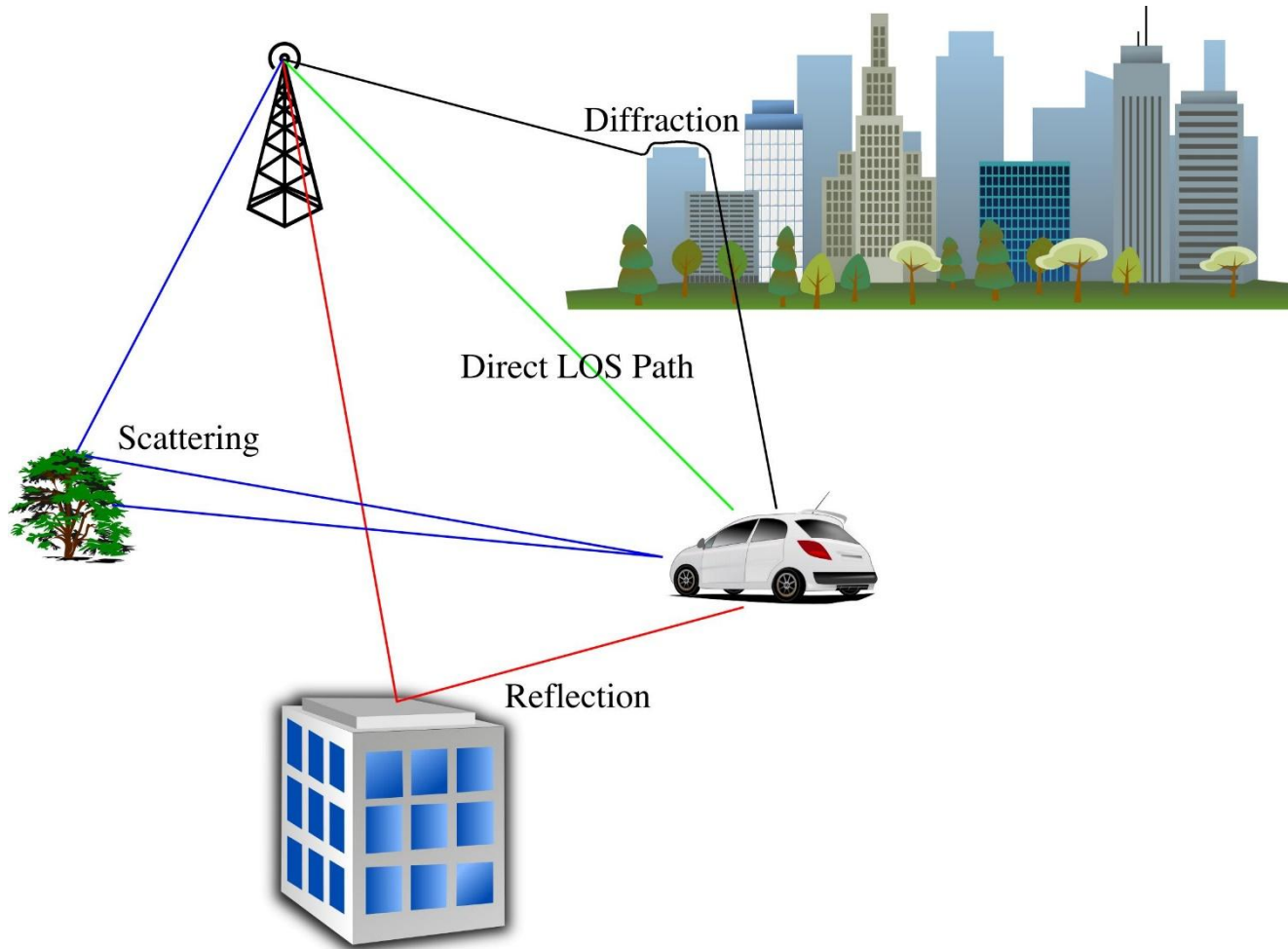
Z Transform: Invertibility



$$\alpha = 0.8, N_0 = 11$$

$$y[n] = x[n] - \alpha x[n - N_0]$$

Z Transform:



$$\alpha = 0.8, N_0 = 11$$

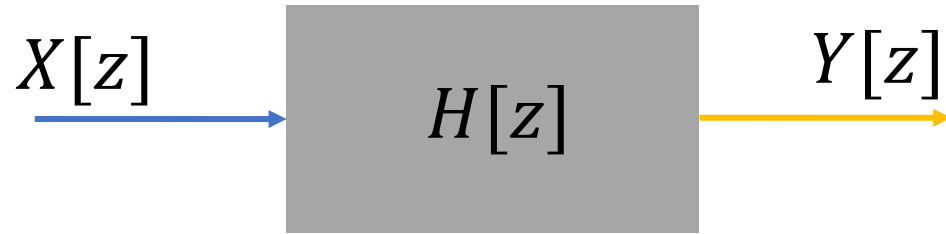
$$y[n] = x[n] - \alpha x[n - N_0]$$

$$Y[z] = X[z] - 0.8Z^{-11}X[z]$$

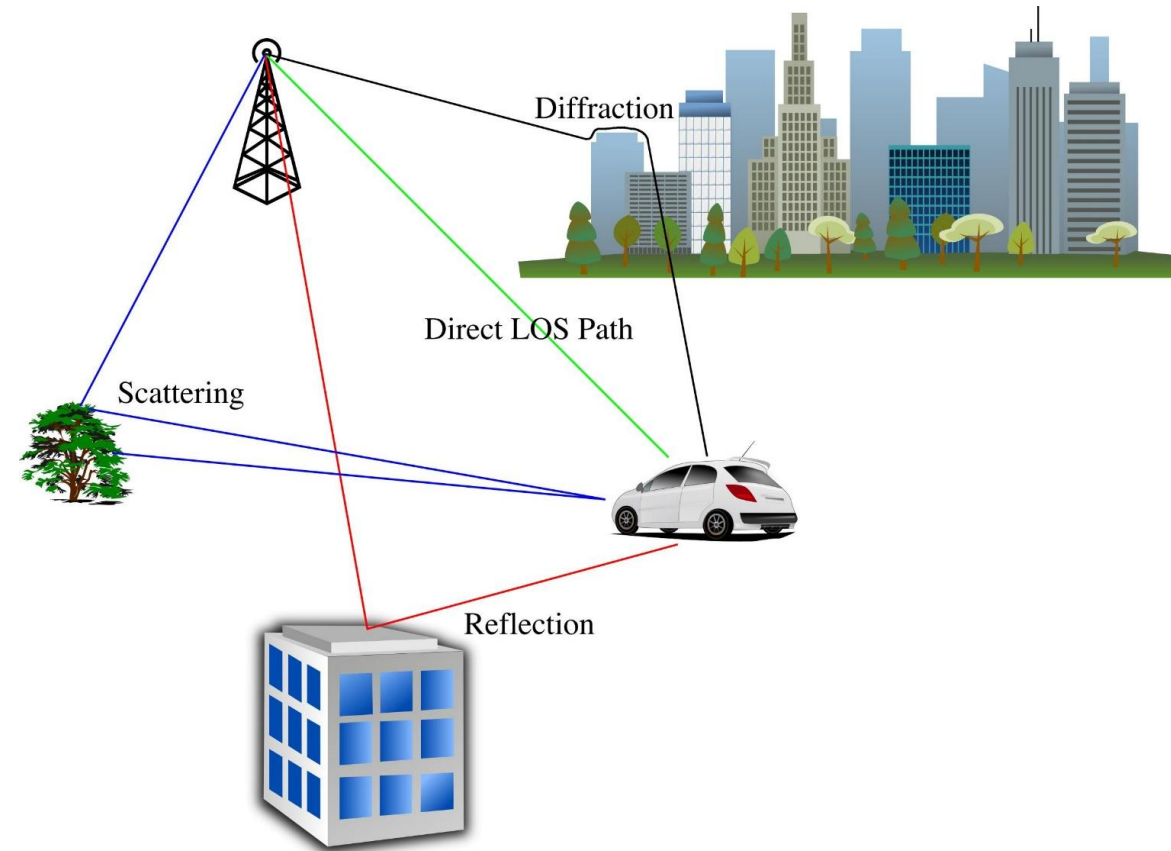
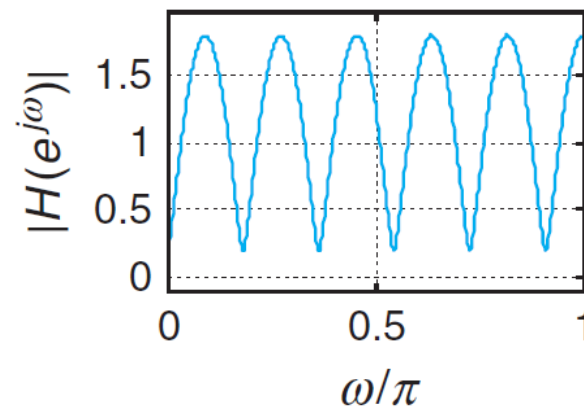
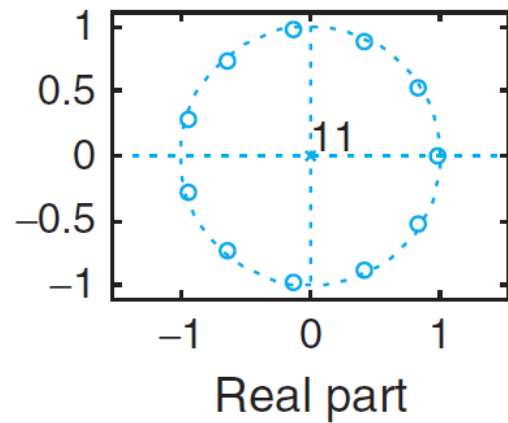
$$Y[z] = (1 - 0.8Z^{-11})X[z]$$

“System” Transfer Function $H[z]$

Z Transform:



$$H(z) = \frac{Y(z)}{X(z)} = 1 - 0.8z^{-11} = \frac{z^{11} - 0.8}{z^{11}}$$

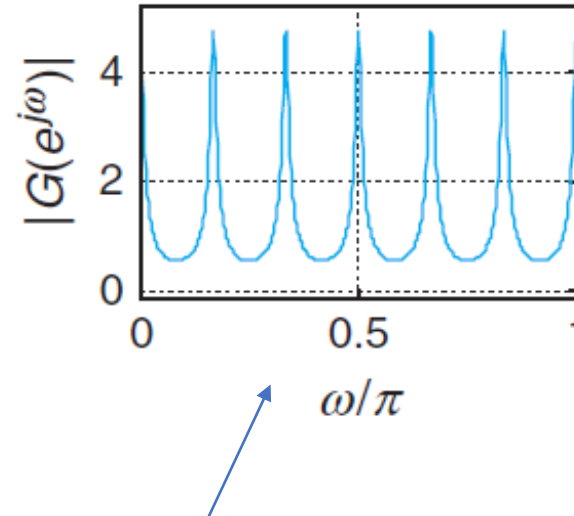
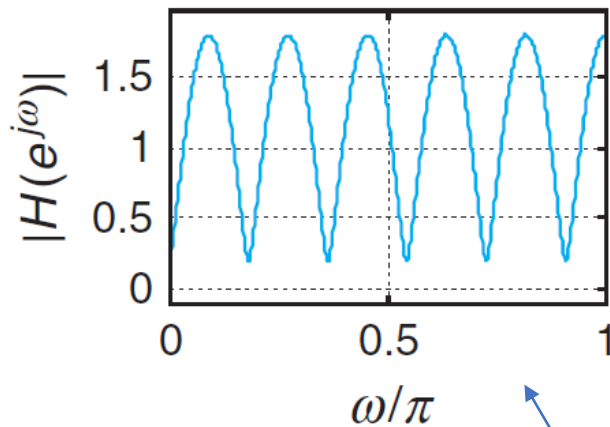


Note that $H[z]$ is invertible (all zeros inside the unit circle), so there is hope.

Z Transform:

The environment behaves as a “comb” filter.
How may we remove this effect to get the original signal back?

One way is to pass the received signal through a filter that cancels the frequency effect of the system



Z Transform:



To get $\hat{X}[z] = X[z]$ we must have $H[z]G[z] = 1$, which gives:

$$G(z) = \frac{z^{11}}{z^{11} - 0.8}$$

Plot Z domain Pole & Zero plot of $G(z)$

This filter will remove the effect of multipath and give our original signal back.



Thanks!